320614(20)

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BE (6th Semester)

Examination, April-May, 2016

(Old Scheme)

Transportation Engineering II

Time Allowed : 3 hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each question is compulsory.

(*ii*) Attempt any **two** parts from (*b*), (*c*) and (*d*).

(*iii*) Distribution of marks – Part (*a*) 2 marks and part (*b*), (*c*) and (*d*) 7 marks each.

(*iv*) Draw neat sketches with proper labelling wherever necessary.

1. (*a*) What are the functions of rail?

(*b*) Write the factors governing the selection of gauge.

(*c*) Describe the purpose and advantage of welding rail.

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(*d*) What are the causes of creep?

2. (*a*) Write the expression for sleeper density for a B.G. track if 17 sleepers are used under a rail length.

(*b*) What are the functions and requirements of good balast?

(*c*) Determine the length of transition curve for design speed of train 95 kmph on a B.G. track.

(*d*) What are the requirements, advantage and disadvantage of metal sleeper?

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3. (*a*) Define facing point of a turnout.

(*b*) What are the types of crossing used in railway? Explain with figures.

(*c*) Calculate all the elements required to set out a 1 in 12 turnout taking off from a straight B.G.

(*d*) What is signalling? Classify different types of signalling.

4. (*a*) Write a short note on tunnel ventillating system requirement.

(*b*) What are commonly adopted tunnel sections?

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(*c*) What are the types of lining used in tunneling?

(*d*) Describe the methods used in hard rock tunneling.

5. (*a*) What are the requirements of a good port?

(*b*) What is breakwater? Describe the types with neat sketches.

(*c*) What are the different types of navigational add?

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(*d*) What is the utility of transit shed and warehouse in a port?

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